PROBLEM STATEMENT

The MSE Compliance Gap

Indian MSEs Are Drowning in BIS Compliance PDFs

- Weeks of manual search through dense BIS SP 21 documents
- No structured discovery tool — only PDFs and generic search
- MSEs lack compliance experts; errors risk legal liability
- Delays slow market entry for Building Material products

MSE Reality Check

63M+	MSEs in India
3–6 wks	Avg. compliance search time
80%	lack in-house compliance teams
~1,000	BIS standards for Building Materials

SOLUTION OVERVIEW

BIS Standards Recommendation Engine

From product description to top-matched BIS standards — in under 5 seconds.

Smart Input

User enters a plain-text product description (e.g., 'OPC 53 Grade Cement')

RAG Retrieval

Hybrid search (semantic + keyword) finds most relevant BIS SP 21 chunks

LLM Reasoning

LLM ranks and explains top 3-5 standards with short. actionable rationales

Instant Output

Streamlit UI delivers results in <5 sec — no expertise needed

✦ Accuracy · Speed · Zero compliance expertise required ✦

SYSTEM ARCHITECTURE

End-to-End RAG Pipeline

01

Data Ingestion

BIS SP 21 PDFs Parsed & extracted

02

Chunking

Semantic sections Overlap preserved

03

Embedding

Vector embeddings generated per chunk

04

Qdrant DB

Vectors stored + indexed for search

05

Retrieval

Hybrid search 0.7 sem / 0.3 kw

06

LLM Output

Ranked standards with rationale

Tech Stack:

- Python
- Qdrant
- Streamlit
- UI.py
- LangChain
- OpenAI Embeddings
- inference.py
- eval_script.py

RETRIEVAL STRATEGY

Chunking + Hybrid Search — Precision by Design

Chunking Strategy

- Semantic sectioning
- Clause-aware boundaries — no mid-sentence splits
- Overlapping windows to preserve context
- Metadata tagged: standard ID, section, page
- Optimal chunk size tuned per BIS document structure

Embedding Generation

- OpenAI text-embedding model (dense vectors)
- Captures semantic meaning beyond keyword overlap
- Vectors stored in Qdrant with payload metadata
- Supports cosine similarity search at scale
- Same model at index + query time for consistency

Hybrid Retrieval

Semantic 70%	KW 30%

- 0.7 semantic weight — captures intent & meaning
- 0.3 keyword weight — anchors exact standard codes
- Best of both: fuzzy + precise matching
- Reranking applied post-retrieval
- Reduces false positives vs. pure semantic search

DEMO RESULTS

Demo Highlights — 10/10 Queries Correct

"33 Grade OrdinaryPortland Cement"

→ **IS 269: 1989** ✓

"Coarse and fine aggregates for structural concrete"

→ **IS 383: 1970** ✓

"Portland pozzolana cement calcined clay based"

→ **IS 1489 (Part 2): 1991** ✓

"Supersulphated cement for marine works"

→ **IS 6909: 1990** ✓

EVALUATION RESULTS

Measured. Validated. Production-Ready.

100%

Hit Rate @3

10 out of 10 queries return the correct standard in top 3

0.90

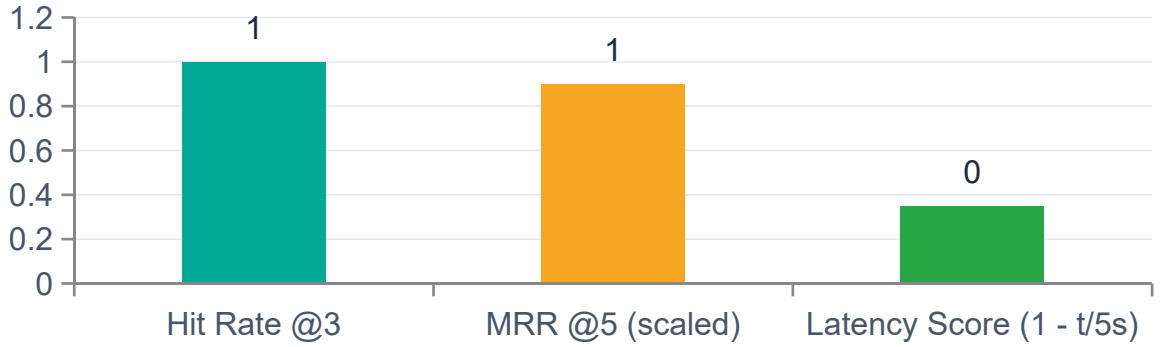
MRR @5

Mean Reciprocal Rank — relevant result appears near top consistently

3.5s

Avg. Latency

Well within the 5-second target — fast enough for real-time use



What the numbers mean

- Hit Rate @3: Correct standard in top 3 — 100% success
- MRR @5: Relevant result ranks high — 0.9 is excellent for retrieval systems
- Latency: 3.5s avg — well under 5s SLA, suitable for real-time UI

REAL-WORLD IMPACT

Transforming MSE Compliance, at Scale

Time Savings

Weeks → Minutes

MSEs identify relevant BIS standards in under 5 seconds vs. manual search taking 3–6 weeks

Cost Reduction

Zero Expert Fees

Eliminates need to hire compliance consultants — saves ₹20,000–₹1L+ per compliance cycle

Faster Market Entry

Accelerated Launch

Compliance bottleneck removed — products reach market faster, revenue realized sooner

Democratized Access

Any MSE, Anywhere

Rural and tier-2 city enterprises gain same compliance intelligence as large corporations

India has 63M+ MSEs. Even 1% adoption = 630,000 businesses with faster, smarter compliance.

TEAM & ACKNOWLEDGEMENTS

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Tools & Libraries

- **Embeddings:** BAAI/bge-small-en-v1.5
- **LLM:** Llama 3.3 70B via Groq LPU
- **Vector DB:** Qdrant (local)
- **Backend:** Python · **Frontend:** Streamlit

Data & Acknowledgements

- Dataset: BIS SP 21 (Bureau of Indian Standards)
- Inspiration: India's MSME Digital India mission
- Hackathon organizers & mentors

"Compliance should empower MSEs — not hold them back. We built the engine to make that real."